

Compressed CO2 (Product)	339.1 K	11000000 Pa	0.002839 kg/s	0.06672 mol/s	0 —	-73.92 kJ/kg	276.6 kg/m ³	Compressor-6 > —
Cool_co2-2	313.1 K	8000000 Pa	0.002839 kg/s	0.06672 mol/s	0 —	-89.08 kJ/kg	236.4 kg/m ³	Cooler-8 > Compressor-6
Comp_co2-2	390.3 K	8000000 Pa	0.002839 kg/s	0.06672 mol/s	0 —	33.96 kJ/kg	124.3 kg/m ³	Compressor-5 > Cooler-8
Comp_co2-1	370.4 K	3700000 Pa	0.002839 kg/s	0.06672 mol/s	0 —	40.2 kJ/kg	56.33 kg/m ³	Compressor-4 > Cooler-7
Comp_H2-1	459.7 K	4500000 Pa	0.0005859 kg/s	0.2774 mol/s	0 —	2241 kJ/kg	2.456 kg/m ³	Compressor-1 > Cooler-3
Off-Gas	308.1 K	1500000 Pa	0.004913 kg/s	0.2032 mol/s	0.003577 —	-6.082 kJ/kg	14.67 kg/m ³	PSA > Cooler-6
Hydrogen	308.1 K	1500000 Pa	0.0005859 kg/s	0.2774 mol/s	0 —	134.3 kJ/kg	1.231 kg/m ³	PSA > Compressor-1
Water	308.1 K	1500000 Pa	0.005612 kg/s	0.311 mol/s	1 —	-2484 kJ/kg	994.6 kg/m ³	G-L Separator-1 > —
Dry Gas	308.1 K	1500000 Pa	0.005499 kg/s	0.4806 mol/s	0 —	19.86 kJ/kg	6.74 kg/m ³	G-L Separator-1 > PSA
Syngas-04	308.1 K	1500000 Pa	0.01111 kg/s	0.7917 mol/s	0.5051 —	-1245 kJ/kg	13.52 kg/m ³	Cooler-2 > G-L Separator-1
Liquid-02	673.1 K	2900000 Pa	0 kg/s	0 mol/s	—	—	—	WGS Reactor > —
Syngas-03	673.1 K	2900000 Pa	0.01111 kg/s	0.7917 mol/s	0 —	916.9 kJ/kg	7.315 kg/m ³	WGS Reactor > Cooler-2
Syngas-02	673.1 K	2900000 Pa	0.01111 kg/s	0.7917 mol/s	0 —	909.7 kJ/kg	7.32 kg/m ³	Cooler-1 > WGS Reactor
Liquid-01	1003 K	3000000 Pa	0 kg/s	0 mol/s	—	—	—	Steam Reformer > —
Syngas-01	1003 K	3000000 Pa	0.01111 kg/s	0.7917 mol/s	0 —	1846 kJ/kg	5.038 kg/m ³	Steam Reformer > Cooler-1
Heated Feed	1003 K	3000000 Pa	0.01111 kg/s	0.6357 mol/s	0 —	1674 kJ/kg	6.298 kg/m ³	Heater-1 > Steam Reformer
Mixed Feed	386.6 K	100000 Pa	0.01111 kg/s	0.6357 mol/s	0 —	174.2 kJ/kg	0.5464 kg/m ³	Mixer > Heater-1
Steam	423.1 K	100000 Pa	0.008333 kg/s	0.4626 mol/s	0 —	232.6 kJ/kg	0.5152 kg/m ³	— > Mixer
Methane	298.1 K	100000 Pa	0.002778 kg/s	0.1732 mol/s	0 —	-1.131 kJ/kg	0.6485 kg/m ³	— > Mixer
Fuel Gas (Reformer)	285.1 K	1500000 Pa	0.002046 kg/s	0.135 mol/s	0.0005923 —	-44.14 kJ/kg	9.821 kg/m ³	Compound Separator > —
CO2 Rich	285.1 K	1500000 Pa	0.002839 kg/s	0.06672 mol/s	0 —	-27.05 kJ/kg	29.66 kg/m ³	Compound Separator > Compressor-4
Cool_co2-1	313.1 K	3700000 Pa	0.002839 kg/s	0.06672 mol/s	0 —	-22.73 kJ/kg	73.33 kg/m ³	Cooler-7 > Compressor-5
Cooler out	285.1 K	1500000 Pa	0.004913 kg/s	0.2032 mol/s	0.005665 —	-46.36 kJ/kg	16.05 kg/m ³	Cooler-6 > G-L Separator-2
Liq-out	285.1 K	1500000 Pa	2.783e-05 kg/s	0.001539 mol/s	1 —	-2577 kJ/kg	998.8 kg/m ³	G-L Separator-2 > —
Vap-out	285.1 K	1500000 Pa	0.004885 kg/s	0.2017 mol/s	0 —	-31.95 kJ/kg	15.96 kg/m ³	G-L Separator-2 > Compound Separator
Comp_H2-3	405.9 K	20000000 Pa	0.0005859 kg/s	0.2774 mol/s	0 —	1529 kJ/kg	11.72 kg/m ³	Compressor-3 > Cooler-5
Compressed H2 (Product)	313.1 K	20000000 Pa	0.0005859 kg/s	0.2774 mol/s	0 —	220.8 kJ/kg	15.04 kg/m ³	Cooler-5 > —

Stream Compositions (mole fractions)

Stream	Methane	Carbon monoxid	Carbon dioxide	Hydrogen	Water
Cool_H2-2	0.006863	0	0	0.9931	0
Comp_H2-2	0.006863	0	0	0.9931	0
Cool_H2-1	0.006863	0	0	0.9931	0
Compressed CO2 (Product)	0.04194	0.006094	0.9474	0.004587	0
Cool_co2-2	0.04194	0.006094	0.9474	0.004587	0
Comp_co2-2	0.04194	0.006094	0.9474	0.004587	0
Comp_co2-1	0.04194	0.006094	0.9474	0.004587	0
Comp_H2-1	0.006863	0	0	0.9931	0
Off-Gas	0.4589	0.02858	0.3535	0.1506	0.008439
Hydrogen	0.006863	0	0	0.9931	0
Water	2.212e-08	3.089e-08	0.001049	4.253e-06	0.9989
Dry Gas	0.198	0.01209	0.1495	0.6368	0.003569
Syngas-04	0.1202	0.007337	0.09115	0.3866	0.3947
Liquid-02	0	0	0	0	0
Syngas-03	0.1202	0.007337	0.09115	0.3866	0.3947
Syngas-02	0.1202	0.03672	0.06177	0.3572	0.424
Liquid-01	0	0	0	0	0
Syngas-01	0.1202	0.03672	0.06177	0.3572	0.424
Heated Feed	0.2724	0	0	0	0.7276
Mixed Feed	0.2724	0	0	0	0.7276
Steam	0	0	0	0	1
Methane	1	0	0	0	0
Fuel Gas (Reformer)	0.6703	0.04002	0.06386	0.2245	0.001339
CO2 Rich	0.04194	0.006094	0.9474	0.004587	0
Cool_co2-1	0.04194	0.006094	0.9474	0.004587	0

Cooler out	0.4589	0.02858	0.3535	0.1506	0.008439
Liq-out	1.201e-08	2.926e-08	0.002812	4.107e-07	0.9972
Vap-out	0.4624	0.0288	0.3561	0.1517	0.0008963
Comp_H2-3	0.006863	0	0	0.9931	0
Compressed H2 (Product)	0.006863	0	0	0.9931	0

Energy Streams

Stream	Energy Flow (kW)
E17	0.8826
E16	0.8837
E15	1.195
E14	0.04305
E13	0.3493
E12	0.161
E7	1.234
E6	0.06043
E5	24.02
E4	-0.8769
E3	10.41
E2	15.95
E1	16.67
E10	0.1909
E8	0.1979
E9	0.01105
E11	0.1787
E18	0.7768
E19	0.7662

Unit Operations

Cooler-4 [Cooler]

Inputs: Comp_H2-2 Outputs: Cool_H2-2

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa ^{0.5}]
Volume	1	m3
Minimum Pressure	1.013	bar
Initialize using Inlet Stream	-1	
Efficiency	100	
Outlet Temperature	40	C
Heat Removed	0.8826	kW
Outlet Molar Vapor Fraction	1	
Temperature Difference	-107.7	C.

Compressor-2 [Compressor]

Inputs: Cool_H2-1, E16 Outputs: Comp_H2-2

Property	Value	Unit
Pressure Increase	55	bar
Adiabatic Efficiency	75	%
Temperature Difference	107.7	C.
Power Required	0.8837	kW
Outlet Pressure	100	bar
Polytropic Efficiency	77.6	%
Adiabatic Coefficient	1.448	
Polytropic Coefficient	1.642	
Adiabatic Head	1.195e+05	m
Polytropic Head	1.237e+05	m
Rotation Speed	1500	rpm
PressureRatio	2.222	

Cooler-3 [Cooler]

Inputs: Comp_H2-1 Outputs: Cool_H2-1

Property	Value	Unit
Flow Conductance	1	[kg/s]/[Pa ^{0.5}]
Volume	1	m3
Minimum Pressure	1.013	bar
Initialize using Inlet Stream	-1	
Efficiency	100	
Outlet Temperature	40	C
Heat Removed	1.195	kW
Outlet Molar Vapor Fraction	1	
Temperature Difference	-146.5	C.

Compressor-6 [Compressor]

Inputs: Cool_co2-2, E14 Outputs: Compressed CO2 (Product)